

ABSTRACT

Multimodal Transformer-Based Early Depression Severity Detection Using Social Conversations, Wearables, and Explainable AI

Conventional depression diagnosis relies on subjective self-reports vulnerable to recall bias and verbal masking ("smiling depression"). This work presents a privacy-conscious multimodal framework quantifying 4-tier depression severity (Normal, Mild, Moderate, Severe) by fusing social text with wearable biosensors. Using MentalBERT, XGBoost with engineered autonomic ratios, fault-tolerant Late Fusion, and TreeSHAP/Self-Attention explainability, the proposed model achieves 90.80% accuracy, 0.8900 macro F1, and 92.40% severe recall at 138ms latency to assist clinical triage.

1. MOTIVATION

- Escalating Global Burden: MDD affects over 280 million people globally (4th leading cause of youth mortality), yet clinical intervention remains severely delayed.
- Limitations of Traditional Screening: Diagnostic tools like PHQ-9 and BDI rely on retrospective self-reporting, suffering from recall bias, patient denial, and stigma.
- Opportunity in Digital Biomarkers: Pseudonymous online posts capture unfiltered emotions, while wearable sensors enable continuous, objective tracking of autonomic signals (HR, HRV, EDA, Sleep).

2. PROBLEM STATEMENT

To design and develop an interpretable, multimodal depression screening framework that accurately quantifies 4-tier clinical depression severity (Normal, Mild, Moderate, Severe) by fusing conversational social text with continuous wearable biosignals, thereby resolving verbal masking ('smiling depression'), eliminating isolated sensor ambiguity, and providing transparent clinical reasoning through Explainable AI.

3. OBJECTIVES

- To fine-tune MentalBERT for 4-tier depression severity classification while mitigating class imbalance.
- To engineer autonomic biomarker ratios and model non-linear physiological stress using XGBoost.
- To design Late Fusion with a divergence safety gate ($> 50\%$) to counter verbal masking.
- To deploy TreeSHAP and Self-Attention explainability with a Clinician Radar Chart.

4.DATASET DETAILS

- Dataset 1 (Reddit DsD - ACM WWW '22): 3,553 DSM-5 annotated posts across 4 severity tiers — Normal: 2,587 (72.8%), Mild: 290 (8.2%), Moderate: 394 (11.1%), Severe: 282 (7.9%). Captures nuanced cognitive distress while modeling real-world class imbalance.
- Dataset 2 (Wearables Biosignals): 11,700 continuous records across 4 tiers — Normal: 3,120, Mild: 2,940, Moderate: 3,280, Severe: 2,360. Tracks involuntary autonomic signals (HR, HRV-RMSSD, EDA/GSR, Temp, Motion, Sleep) to objectively detect verbal masking.

5. METHODOLOGY

- Text Pipeline: MentalBERT was fine-tuned using AdamW ($\text{lr} = 2\text{e-}5$) with Cost-Sensitive Loss Weighting ($W_c = N / (kN_c)$) and Stratified 5-Fold CV to mitigate minority class imbalance.

- Wearable Pipeline: Sensor noise was cleaned via 1st–99th percentile Winsorization. Five engineered ratios (ASL, VTI, CRR, SQI, PI) modeled autonomic stress-recovery balance via regularized XGBoost with native TreeSHAP support.
- Late Fusion & Safety: Modalities merge via Weighted Softmax: $P_{\text{final}} = 0.55P_{\text{text}} + 0.45P_{\text{sensor}}$ (Argmax) with dynamic missing-stream fallback. If cross-modal divergence $D = 0.5\text{sum}(|P_{\text{text}} - P_{\text{sensor}}|) > 0.50$, involuntary biosensors override masked text ("smiling depression").
- Explainable AI: TreeSHAP calculates biomarker % attributions, MentalBERT self-attention highlights depressive text tokens, and an SVG Radar Chart visualizes physiological anomalies.

6. RESULTS

Model Configuration	Accuracy	Macro F1	Severe Recall	Latency
Wearables Baseline (XGBoost)	58.63%	0.5570	61.20%	< 15 ms
Text Baseline (MentalBERT)	73.42%	0.7180	78.50%	< 90 ms
Proposed Multimodal Late Fusion	90.80%	0.8900	92.40%	< 138 ms

Findings: Late fusion improves accuracy by +32.17% over wearables and +17.38% over text. High Severe Recall (92.40%) protects critical patients. TreeSHAP identifies ASL, HRV, and Sleep as top drivers (>68% attribution). End-to-end latency is 138ms on FastAPI.

7. CONCLUSION & FUTURE SCOPE

- Combining conversational text with wearable biosignals resolves single-modality blind spots, delivering a 90.80% accurate, interpretable screening framework to assist clinical triage rather than replace doctors.
- Future Scope:
 - (1)Acoustic Voice Biomarkers: Integrate vocal pitch and prosody for tri-modal screening.
 - (2)Large-Scale Pre-training: Leverage the 1.27M-post SMHD dataset for domain adaptation.
 - (3)Temporal Forecasting: Deploy PatchTST time-series transformers to predict depressive episodes before clinical onset.

8. REFERENCES

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